

Course Code: CS240

Max Marks: 80

Subject: Operating Systems & Distributed Computing

Time: 3 Hours

University of Stanford - Department of Computer Science

N.B.: (1) Question No. 1 is Compulsory.

(2) Attempt any three questions out of remaining five.

Q1. Attempt any FIVE of the following:

- a) Explain race conditions and mutual exclusion using Peterson's algorithm. [4]
- b) Discuss virtual memory management and page replacement algorithms. [4]
- c) Explain process scheduling algorithms in multi-core operating systems. [4]
- d) Describe deadlock prevention, avoidance, and Banker's algorithm. [4]
- e) Explain distributed file systems architecture and consistency models. [4]
- f) Discuss kernel microarchitecture vs monolithic kernels. [4]

Q2. a) Explain the structure of file system inodes and block allocation methods. [10]

b) Discuss semaphores and condition variables with bounded buffer problem. [10]

Q3. a) Explain memory paging and segmentation mechanisms in x86 hardware. [10]

b) Discuss Raft consensus algorithm for distributed state machines. [10]

Q4. a) Explain CPU scheduling algorithms including FCFS, SJF, and Round Robin. [10]

b) Discuss thrashing and working set model in memory management. [10]

Q5. a) Explain two-phase commit protocol for distributed transactions. [10]

b) Discuss I/O subsystem architecture and disk scheduling algorithms. [10]

Q6. a) Explain virtual machine monitors and hypervisor architectures. [10]

b) Discuss security mechanisms in operating systems and access control lists. [10]